

PRD-005 — Personalized Practice

Product: PrepHub V2

Status: Approved

Depends on:

- PRD-003 — Billing & Subscriptions
- PRD-004 — Student Home Dashboard
- PRD-006 — Practice Session
- PRD-007 — Session Review & Results
- PRD-012 — Diagnostic Assessment
- PRD-013 — Question Content System

Related Document:

- PRD-014 — Adaptive Recommendation Engine
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1. Purpose

Personalized Practice is PrepHub's core recurring learning experience.

Students complete automatically generated 21-question practice sets that adapt to their demonstrated performance.

PrepHub determines what the student should practice. Students do not manually select categories, difficulties, or question counts.

The Practice page acts as the entry point between the Student Home Dashboard and the active question-by-question Practice Session.

2. Product Philosophy

PrepHub should remove the burden of deciding what to study.

The student's responsibility is to:

- Try their best
- Complete the generated practice set
- Review immediate explanations
- Continue practicing consistently

PrepHub's responsibility is to:

- Select appropriate questions
- Preserve unfinished work
- Update the student's performance model
- Generate the next personalized set
- Clearly guide the student toward the next action

The practice workflow should feel structured and predictable, while the question selection remains personalized and varied.

3. Goals

Personalized Practice must:

- Provide automatically generated adaptive practice sets.
 - Present each set as a clear, finishable unit.
 - Allow students to pause and resume unfinished sets.
 - Show current set progress before re-entry.
 - Prevent students from discarding or restarting generated sets.
 - Provide immediate learning feedback after every submitted answer.
 - Update the PrepHub Score Prediction after each completed set.
 - Generate the next set automatically.
 - Keep practice unavailable until the diagnostic is completed.
 - Keep paid practice unavailable without active individual or school-sponsored access.
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4. Non-Goals

PRD-005 does not define:

- The mathematical adaptive algorithm
- Category Ability Score calculations
- Category priority formulas

- Difficulty distributions
- Exact-question eligibility rules
- Question-family cooldown formulas
- Score-prediction calculations
- Question-answering interface details
- Session-results interface details

Those requirements are defined in PRD-006, PRD-007, and PRD-014.

Personalized Practice does not allow students to:

- Choose categories
- Choose difficulty
- Select a custom number of questions
- Build custom practice sets
- Delete an active set
- Restart an active set
- Skip directly to a newly generated set
- Permanently browse old practice questions

5. Practice Eligibility

A student may begin Personalized Practice only when:

1. The initial diagnostic has been completed.
2. The student has active access through either:
 - An individual subscription, or
 - A partnered school or district.

Students who have not completed the diagnostic are redirected to begin or resume it.

Students who completed the free diagnostic but do not have active paid or school-sponsored access may view their diagnostic results but cannot enter the first adaptive practice set.

6. First Adaptive Set

Completing the diagnostic automatically generates the student's first personalized adaptive set.

The set is generated even when the student does not yet have paid or school-sponsored access.

For an unpaid student:

- The generated set remains saved.
- The Practice page remains locked.
- The student is directed toward subscription options.
- The set becomes available when valid access begins.

The student does not need to retake the diagnostic or regenerate the initial set after subscribing.

7. Practice Navigation

Students can reach the Practice page from:

- The main navigation
- The Student Home Dashboard's **Continue Practice** button
- The Session Review flow
- Other approved practice calls to action

The Dashboard's **Continue Practice** button routes to the Practice page rather than directly opening a question.

This keeps the Dashboard focused on progress and makes the Practice page the consistent starting point for all practice sessions.

8. Practice Page

The Practice page displays the student's current or next adaptive set.

The page should remain simple and focused.

It includes:

- Practice set number
- Total number of questions
- Current completion progress
- Personalized explanation

- Primary practice action

Example:

Practice Set 8

21 Questions

Progress: 9 of 21 completed

Personalized from your performance.

Primary button:

Continue Practice

For a new set that has not yet begun, the button may read:

Start Practice

Selecting the primary action immediately opens the active Practice Session.

9. Practice Set Numbering

Every adaptive set receives a sequential number for that student.

Examples:

- Practice Set 1
- Practice Set 2
- Practice Set 3

The initial diagnostic is not counted as Practice Set 1.

The first post-diagnostic adaptive set is Practice Set 1.

Set numbering exists to give students a sense of progress and continuity.

Set numbers do not imply that every student receives the same questions.

10. Practice Set Size

Every adaptive practice set contains exactly **21 questions**.

Students cannot change the number of questions.

The fixed set size provides:

- A clear completion goal
- Consistent score updates
- Stable adaptive evaluation
- Predictable session structure
- A meaningful completion reward

The exact category, difficulty, and question composition is determined by PRD-014.

11. One Active Set

A student may have only one active adaptive set at a time.

A set is considered active after it has been generated and before all 21 questions have been completed.

If an active set exists:

- The Practice page displays its progress.
- **Continue Practice** resumes that exact set.
- A new set cannot be generated.
- The student cannot discard or replace it.
- The student cannot restart it.

This prevents students from avoiding questions they do not want to answer and protects the integrity of the adaptive system.

12. Set States

Every adaptive practice set may be in one of the following states:

- Generated
- Not Started
- In Progress
- Completed
- Under Review
- Review Exited

Generated

The set exists but has not yet been made available or opened.

This may apply to unpaid students whose first set was generated after the free diagnostic.

Not Started

The student has access, but no question in the set has been submitted.

In Progress

The student has begun the set and has not completed all 21 questions.

Completed

All 21 questions have been submitted.

Under Review

The student is viewing the Session Review & Results experience.

Review Exited

The student has left the Session Review, and the next adaptive set is available from the Practice page.

13. Starting a New Set

When a new adaptive set is ready, the Practice page displays:

- Set number
- 21-question length
- The message:
Personalized from your performance.
- A **Start Practice** button

Selecting **Start Practice** opens Question 1.

The student is not shown:

- Category allocation
- Difficulty distribution
- Ability calculations
- Priority scores
- Why any specific question was selected

The adaptive engine remains behind the scenes.

14. Resuming an Existing Set

When a student returns to an unfinished set, the Practice page displays:

- Set number
- Questions completed
- Questions remaining
- Visual progress
- **Continue Practice**

Example:

Practice Set 8

9 of 21 questions completed

Personalized from your performance.

Continue Practice

Selecting the button immediately returns the student to the next appropriate unanswered question.

PrepHub should not require the student to manually locate where they stopped.

15. Leaving a Set

Students may voluntarily leave a practice set at any time.

Leaving the Practice Session:

- Does not delete the set
- Does not submit unanswered questions
- Does not generate another set
- Does not reset progress
- Does not affect the student's score merely because they left

The student may return through either the Dashboard or Practice page.

A confirmation warning is not required during normal navigation because all work is autosaved.

If an action could genuinely cause unsaved data loss, PrepHub must warn the student, but the intended implementation should prevent such loss.

16. Autosave and Persistence

PrepHub automatically saves all meaningful practice state.

Saved data includes:

- Active set ID
- Set number
- Question order
- Current position
- Submitted answers
- Submitted correctness
- Unsubmitted selected answers
- Skipped questions
- Questions already reviewed
- Completion progress
- Set status

If the browser closes, the device shuts down, or the student signs in on another supported device, the active set should resume correctly.

The order and membership of the generated set must remain unchanged.

17. Active-Set Content Stability

Once a student begins a generated practice set, all questions in that set remain part of the session.

If the owner later:

- Edits a question
- Unpublishes a question
- Deletes a question from the active bank

the question remains preserved inside the student's already-started set.

The active session should retain the question content required to complete and review that session.

Changes to the question bank affect future generated sets, not an already-started set.

18. Question Navigation

Within the active Practice Session, students may:

- Move among unanswered questions
- Skip a question temporarily
- Return to skipped questions
- Review previously submitted questions when allowed by PRD-006
- Continue from the next unanswered question

Students cannot change an answer after submitting it.

The detailed question interface is defined in PRD-006.

19. Question Submission and Immediate Feedback

When a student deliberately submits an answer:

- The response becomes final.
- Correctness is recorded.
- The student immediately sees whether it was correct.
- The correct answer is displayed.
- The required video explanation is available.
- An optional written explanation is displayed when available.

The student controls when to advance using the **Next Question** button.

The overall PrepHub Score Prediction does not update after each individual question.

It updates only after the entire set is completed.

20. Suggested Time

Each question contains its own suggested-time value.

The Practice Session displays this guidance.

If the suggested time is exceeded:

- PrepHub informs the student.
 - The student may continue.
 - The question is not automatically submitted.
 - The answer may still receive full credit.
 - No direct penalty is applied.
 - Response time does not directly influence the adaptive engine in V1.
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21. Completing a Set

A set is completed only when all 21 questions have been submitted.

If unanswered questions remain and the student attempts to finish, PrepHub displays a reminder and directs them back to those questions.

The set cannot be completed with blanks.

Unlike temporary skipping during the session, final completion requires a submitted response for every question.

22. Completion Processing

Immediately after the 21st required answer is submitted, PrepHub begins completion processing.

The system:

1. Finalizes the set's results.
2. Updates category performance.
3. Updates the PrepHub Score Prediction.
4. Records study statistics.
5. Updates relevant streak and milestone data.
6. Generates the next adaptive practice set.
7. Opens the Session Review & Results experience.

The exact adaptive logic and score calculations are defined in PRD-014 and the applicable score-prediction specification.

23. Next-Set Generation Timing

The next adaptive practice set is generated immediately after the current set is completed.

Generation occurs in the background while the student enters Session Review.

This is intended to ensure that the next set is already available when the student exits the review experience.

The student must not be forced to wait unnecessarily before continuing practice.

24. Session Review Exit

At the end of Session Review, students receive:

Primary action:

Continue Practice

Secondary action:

Back to Dashboard

Selecting **Continue Practice** does not open the first question immediately.

It returns the student to the Practice page.

The Practice page then displays the new set number, 21-question length, personalization message, and **Start Practice** button.

This preserves the Practice page as the consistent gateway into every adaptive set.

25. Previously Completed Questions

Students may review completed questions:

- Immediately after answering each question
- During the completed Session Review experience

After leaving Session Review, individual questions from that completed set are no longer available through a permanent question archive.

PrepHub continues storing the necessary result and analytics data.

The student retains access to long-term progress information but not an unlimited library of previously completed question content.

26. Practice Access Without Subscription

An unpaid independent student who completed the diagnostic may reach the Practice page but cannot begin the generated adaptive set.

The page should display:

- That a personalized set is ready
- That active access is required
- A clear path to view subscription plans

Example:

Your first personalized practice set is ready.

Subscribe to continue with 21 questions selected from your diagnostic performance.

Primary action:

View Plans

The questions themselves must remain hidden until access begins.

27. Practice Loading State

When the Practice page is retrieving or generating set data, display a clear, calm loading state.

Example:

Preparing your personalized practice set...

PrepHub is selecting questions based on your latest performance.

Avoid technical language, algorithm details, or fake progress percentages.

28. Set-Generation Error

PRD-014 should contain fallback logic for limited eligible content.

However, if a complete set still cannot be generated, the student must receive a friendly error state.

Recommended message:

We're having trouble preparing your next practice set.

Your progress is safe. Please try again in a moment.

Actions:

- **Try Again**
- **Back to Dashboard**

The error must not:

- Delete the completed session
- Change the student's updated score
- Generate duplicate active sets
- Expose internal content-bank details

The owner system should receive an internal alert containing enough information to diagnose the failure.

29. Empty and Exceptional States

Diagnostic Incomplete

Complete your diagnostic so PrepHub can personalize your practice.

Action:

Begin Diagnostic or **Resume Diagnostic**

Access Required

Your personalized practice set is ready. Choose a plan to continue.

Action:

View Plans

Active Set

Display progress and **Continue Practice**.

New Set Ready

Display set number, 21 questions, personalization message, and **Start Practice**.

Generation Failure

Display the approved friendly error state and retry option.

30. Responsive and Accessibility Requirements

The Practice page follows PRD-000 and must be:

- Desktop-first
- Fully responsive
- Keyboard accessible
- Screen-reader compatible
- Easy to understand without color alone
- Free of unnecessary visual clutter

Set progress must remain readable across desktop, tablet, and mobile.

The primary practice action must always be visually clear.

31. Acceptance Criteria

Eligibility

- Practice remains unavailable until the diagnostic is completed.
- Practice requires active individual or school-sponsored access.
- The first adaptive set is generated after the diagnostic even for unpaid students.
- Unpaid students cannot view or enter the generated questions.

Practice Page

- Displays the current or next set number.
- Displays that each set contains 21 questions.
- Displays current progress for an unfinished set.
- Displays the message **Personalized from your performance.**
- Includes a Start Practice or Continue Practice button.
- Dashboard and main navigation both route students to the Practice page.

Set Management

- Each student may have only one active set.
- Students cannot delete an active set.
- Students cannot restart an active set.
- Students cannot generate another set before finishing the current one.
- Existing active-set questions remain preserved if their source content is later edited, unpublished, or deleted.

Persistence

- Students may leave and resume later.
- Autosave preserves all required practice state.
- Continue Practice resumes at the next appropriate unanswered question.
- The set's question order and membership remain stable.

Question Flow

- Submitted answers are final.
- Immediate correctness and explanations appear after submission.
- Suggested-time expiration does not force submission.
- Students may skip temporarily.
- All 21 questions must ultimately be submitted.

Completion

- The completed set updates category performance.
- The completed set updates the PrepHub Score Prediction.
- The next adaptive set begins generating immediately.
- Session Review opens after completion.
- Administrator-defined analytics and learning metrics are recorded.

Next Set

- The next set is ready when possible before Session Review ends.
- Continue Practice from Session Review returns to the Practice page.
- The Practice page then allows the student to start the new set.

Review

- Students may review questions during immediate feedback and Session Review.
- Completed questions are not permanently accessible after exiting Session Review.
- Long-term progress remains available.

Error Handling

- A friendly loading state is shown during generation.
- Generation failure preserves all completed progress.
- Students may retry or return to the Dashboard.
- Duplicate active sets are not created.
- Internal failure details are surfaced to the owner rather than the student.

Overall Experience

The student should always understand:

- Which practice set they are on
- How many questions the set contains
- How much of the current set is complete
- Whether they should start or continue
- That the questions were personalized from their performance
- That they cannot discard an unfinished set
- What happens after completing all 21 questions